SPARC Pilot Test of Structural Dark-Matter Residuals

A MAAT v1.6 Real-Data Bridge from Rotation-Curve Residuals to NFW/MOND Baselines

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Abstract

The preceding MAAT dark-matter note formulated galactic dark-matter residuals as structural coherence diagnostics, but only on a synthetic rotation curve. That result was intentionally weak. Paper 61 closes the next methodological gap by running the diagnostic on the SPARC rotation-curve data of 175 disk galaxies.

The pipeline computes baryonic contributions, residual velocity components, MAAT support profiles $(H, B, S_{\rm eff}, V)$, the robustness closure $R_{\rm rob}$, and the structural residual score

$$D_{\rm struct}(r) = f_{\rm res}(r)\, R_{\rm rob}(r)\, V(r).$$

It compares galaxy-level $D_{\rm struct}$ against an NFW-like residual mismatch, a MOND/RAR acceleration mismatch, and shuffled-galaxy null models.

Using fixed mass-to-light factors $\Upsilon_{\rm disk} = 0.5$ and $\Upsilon_{\rm bulge} = 0.7$, the SPARC run contains 3391 radius points. The mean residual fraction is 0.5592, the mean structural residual support is 0.1652, and the mean robustness is 0.5814. At galaxy level, $D_{\rm struct}$ anticorrelates with both NFW-like mismatch ($\rho_{\rm S} = -0.6378$) and RAR mismatch ($\rho_{\rm S} = -0.3256$), while shuffled-galaxy null averages are only $|\rho_{\rm S}| \simeq 0.06$. This is not a dark-matter solution or a precision halo fit, but it is the first real-data indication that the structural residual diagnostic carries nontrivial SPARC-level information.

1 Purpose

The purpose of Paper 61 is not to claim that MAAT explains dark matter. Its purpose is narrower:

> *turn the previous conceptual rotation-residual diagnostic into a predeclared real-data pilot test.*

The previous note was deliberately synthetic. Its main criticism was obvious: it renamed the familiar residual $v_{\rm obs}^2 - v_{\rm bar}^2$ in MAAT language. A nontrivial step requires real galaxy data and external baselines. SPARC is the natural first target because it provides rotation curves and baryonic mass models for 175 disk galaxies [1].

2 SPARC Data Model

The expected local input table has one row per galaxy radius and contains

$$\{G_i, r, v_{\rm obs}, \sigma_v, v_{\rm gas}, v_{\rm disk}, v_{\rm bulge}\}. \tag{1}$$

For the present run, the table is prepared from the Zenodo-hosted `Rotmod_LTG.zip` SPARC rotation-curve files [1]. The repository includes attribution and license notes for these data.

The baryonic contribution is computed as

$$\boxed{v_{\rm bar}^2(r) = v_{\rm gas}|v_{\rm gas}| + \Upsilon_{\rm disk} v_{\rm disk}|v_{\rm disk}| + \Upsilon_{\rm bulge} v_{\rm bulge}|v_{\rm bulge}|.} \tag{2}$$

The signed-square convention allows SPARC-style signed gas contributions while remaining operational for ordinary nonnegative velocity components.

The residual velocity component is

$$\boxed{v_{\rm res}^2(r) = \max\left[v_{\rm obs}^2(r) - v_{\rm bar}^2(r), 0\right],} \tag{3}$$

with residual fraction

$$f_{\rm res}(r) = \frac{v_{\rm res}^2(r)}{v_{\rm obs}^2(r)}. \tag{4}$$

3 MAAT Diagnostic

The support definitions follow the previous dark-residual note but are written as a real-data protocol.

- $H(r)$ penalises baryonic overshoot, where $v_{\rm bar}^2 > v_{\rm obs}^2$ would make the residual construction physically strained.
- $B(r)$ measures radial smoothness of the observed dynamical trajectory.
- $S_{\rm eff}(r)$ measures controlled activity in the residual transition profile.
- $V(r)$ measures connectedness between the baryonic-density proxy and the residual-density proxy.

The closure is

$$R_{\rm resp}(r) = [H(r)B(r)V(r)]^{1/3}, \tag{5}$$

$$R_{\rm rob}(r) = \min\Big[R_{\rm resp}(r), (H(r)B(r)S_{\rm eff}(r)V(r))^{1/4}\Big]. \tag{6}$$

The structural dark-residual diagnostic is

$$\boxed{D_{\rm struct}(r) = f_{\rm res}(r)\, R_{\rm rob}(r)\, V(r).} \tag{7}$$

This definition remains intentionally conservative. A large residual alone is not enough; the residual must also be robustly connected to the baryonic and dynamical structure.

4 External Baselines

4.1 NFW-Like Halo Channel

The code implements a lightweight NFW-like residual-shape fit. For a scale radius r_s, the residual shape is

$$u_{\rm NFW}(r; r_s) = \frac{\ln(1 + r/r_s) - (r/r_s)/(1 + r/r_s)}{r}. \tag{8}$$

For each r_s, an amplitude is fitted by weighted least squares to $v_{\rm res}^2(r)$, and the best r_s is selected on a grid. This is not a full halo inference pipeline, but it provides a reproducible NFW-like comparison channel. Future real-data runs should compare against published SPARC halo catalogues such as Li et al. [6].

4.2 MOND/RAR Channel

The acceleration baseline uses

$$g_{\mathrm{obs}}(r) = \frac{v_{\mathrm{obs}}^2(r)}{r}, \qquad g_{\mathrm{bar}}(r) = \frac{v_{\mathrm{bar}}^2(r)}{r}. \tag{9}$$

A simple radial-acceleration-relation proxy is

$$g_{\mathrm{RAR}}(r) = \frac{g_{\mathrm{bar}}(r)}{1 - \exp[-\sqrt{g_{\mathrm{bar}}(r)/g_\dagger}]}, \qquad g_\dagger \simeq 1.2 \times 10^{-10}\,\mathrm{m\,s^{-2}}, \tag{10}$$

following the spirit of the observed radial acceleration relation [5]. The resulting velocity is compared with v_{obs} in units of the observational velocity uncertainty.

5 Null Tests

The first predeclared null test is a shuffled-galaxy null. Galaxy-level summary values of D_{struct} are randomly permuted before computing correlations with NFW-like and RAR mismatch scores. A useful positive result would require the real pairing to exceed the shuffled null robustly.

A stronger follow-up run should also include:

- shuffled-radius nulls within each galaxy;
- family-stratified tests for low-surface-brightness, high-surface-brightness, dwarf, and bulge-dominated systems;
- leave-family-out tests to see whether support definitions generalise;
- comparison with published multi-halo fits, not only the lightweight NFW-like proxy.

6 SPARC Pilot Run

The real SPARC run uses the prepared `Rotmod_LTG` rotation-curve data. It contains 175 galaxies and 3391 radial measurements. The present pilot uses fixed mass-to-light factors rather than fitting stellar population parameters, so it should be read as a diagnostic benchmark rather than a precision dark-matter inference.

Table 1: Paper 61 SPARC pilot summary.

Quantity	**Value**
mode	SPARC real input
galaxies	175
radius rows	3391
mean residual fraction	0.5592
mean D_{struct}	0.1652
mean R_{rob}	0.5814
mean V	0.4661
Spearman D_{struct} vs NFW-like mismatch	−0.6378
Spearman D_{struct} vs RAR mismatch	−0.3256
mean shuffled-null $\lvert\rho\rvert$ for NFW channel	0.0608
mean shuffled-null $\lvert\rho\rvert$ for RAR channel	0.0593

The negative sign is physically favourable in this setup: higher structural support is associated with smaller mismatch relative to the external baseline. Both real absolute correlations exceed shuffled-galaxy null averages by a wide margin. This is the first nontrivial real-data signal for the diagnostic, but it remains a pilot result because the NFW-like fit is lightweight and the stellar mass-to-light factors are fixed.

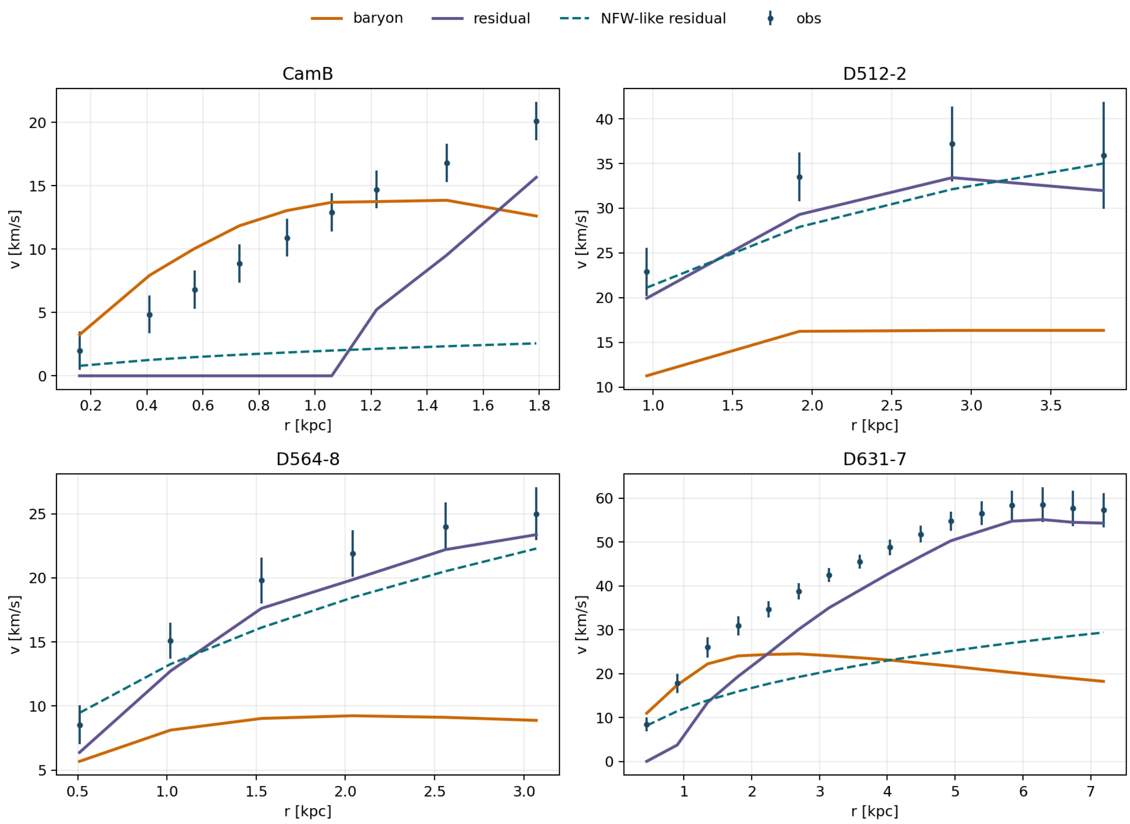

Figure 1: SPARC rotation-decomposition examples. The figure shows observed velocities, baryonic contributions, inferred residuals, and NFW-like residual fits for representative galaxies.

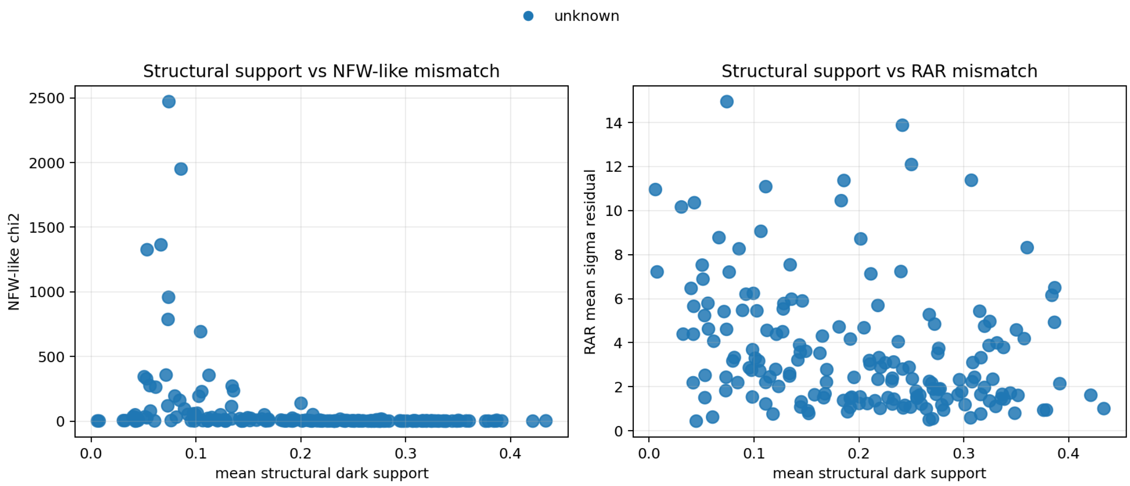

Figure 2: SPARC population-level structural support versus NFW-like and RAR mismatch. The negative trend indicates that larger structural residual support is associated with smaller baseline mismatch in this pilot setup.

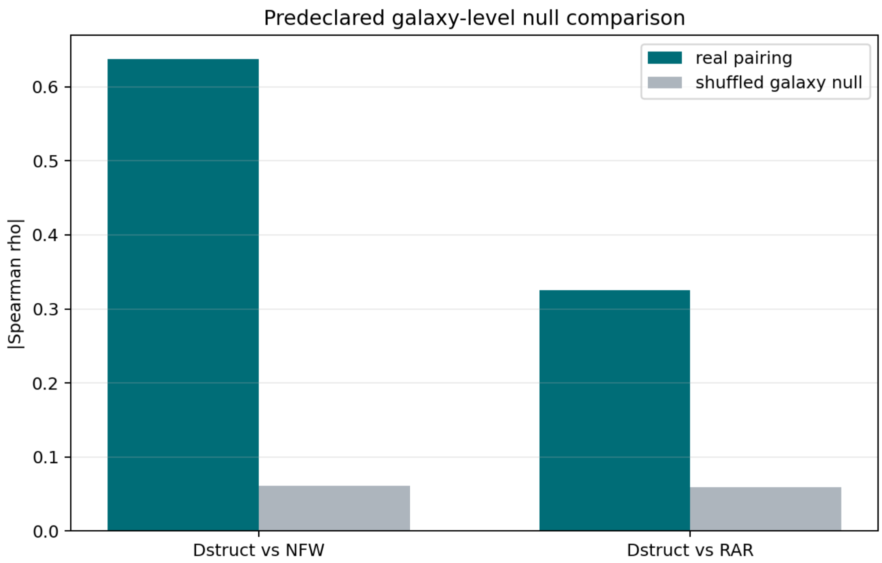

Figure 3: Shuffled-galaxy null comparison. The real SPARC pairings exceed shuffled-null averages in absolute Spearman correlation.

7 Interpretation

Paper 61 is best understood as a first real-data bridge paper. It moves the dark-residual idea from a synthetic explanatory note to an executable SPARC benchmark.

The most important methodological point is the failure mode:

> *If D_{struct} only tracks raw residual fraction or baryonic acceleration and does not beat shuffled or halo/RAR baselines, then MAAT adds no dark-matter information in this setting.*

The present run does not fail this first test: the real SPARC pairing is much stronger than the shuffled-galaxy null. However, the result should be interpreted cautiously. The NFW channel is not a full halo inference, the RAR channel is a simple proxy, and stellar mass-to-light factors are fixed. The correct conclusion is therefore:

> *MAAT structural residual support carries measurable SPARC-level signal in this pilot, but it has not yet been shown to outperform precision halo or MOND/RAR analyses.*

8 Reproducibility

Repository:

https://github.com/Chris4081/structural-selection-principle

Experiment folder:

```
experiments/sparc_structural_residual_pilot_paper61/
```

Prepare SPARC Rotmod data:

```
python3 prepare_sparc_rotmod.py
```

Mock schema test:

```
python3 sparc_structural_residual_pilot.py --mode mock
```

SPARC real-data run:

```
python3 sparc_structural_residual_pilot.py \
  --input data/sparc_rotation_curves.csv \
  --output outputs_sparc_real
```

The script writes:

- `paper61_radius_diagnostics.csv`
- `paper61_galaxy_summary.csv`
- `paper61_summary.json`
- `fig1_rotation_decomposition_examples.png`
- `fig2_population_structural_vs_baselines.png`
- `fig3_shuffled_galaxy_nulls.png`

9 Data Attribution and License Notes

The SPARC rotation-curve data used here were obtained from the Zenodo-hosted SPARC record. The Zenodo record lists the license as Creative Commons Attribution 4.0 International (CC-BY-4.0). The repository includes the raw `Rotmod_LTG.zip`, a prepared CSV, and a local attribution note.

When using or redistributing these data:

- cite the original SPARC paper by Lelli, McGaugh, and Schombert [1];
- cite the Zenodo DOI 10.5281/zenodo.16284118 when using the Zenodo-hosted record;
- respect the Zenodo record's CC-BY-4.0 attribution requirement;
- include the standard VizieR/CDS acknowledgment if using the VizieR catalogue route;
- do not imply endorsement by the SPARC authors, Zenodo, VizieR, CDS, or the original data providers;
- clearly distinguish original SPARC measurements from derived MAAT diagnostic artifacts.

10 Scope and Compliance Checklist

For clarity, the status of the present pilot is:

- **Data attribution:** SPARC is attributed to Lelli, McGaugh, and Schombert, with the Zenodo DOI and CC-BY-4.0 notice included.
- **No endorsement:** no endorsement by the SPARC authors, Zenodo, VizieR, CDS, or the original data providers is claimed or implied.
- **Negative framing allowed:** this is a pilot correlation study, not a dark-matter model fit and not evidence that MAAT explains dark matter.
- **Baseline separation:** the NFW-like and MOND/RAR-like channels are comparison diagnostics only. They are not full professional halo fits, population-inference pipelines, or precision modified-dynamics analyses.

11 Conclusion

Paper 61 provides the first SPARC real-data pilot for MAAT dark-residual diagnostics. The result is modest but nontrivial: D_{struct} anticorrelates with NFW-like and RAR mismatch, and the real galaxy pairing is stronger than shuffled-galaxy nulls.

This should not be interpreted as a solution to dark matter. It is evidence that the structural residual diagnostic is not merely a synthetic toy: it carries measurable information on real SPARC rotation curves. The next step is to replace the lightweight baselines by published halo catalogues, fitted stellar mass-to-light factors, lensing where available, and family-stratified out-of-sample tests.

References

[1] F. Lelli, S. S. McGaugh, and J. M. Schombert, "SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves," *Astronomical Journal* **152**, 157 (2016).

[2] J. F. Navarro, C. S. Frenk, and S. D. M. White, "The Structure of Cold Dark Matter Halos," *Astrophysical Journal* **462**, 563–575 (1996).

[3] J. F. Navarro, C. S. Frenk, and S. D. M. White, "A Universal Density Profile from Hierarchical Clustering," *Astrophysical Journal* **490**, 493–508 (1997).

[4] M. Milgrom, "A modification of the Newtonian dynamics as a possible alternative to the hidden mass hypothesis," *Astrophysical Journal* **270**, 365–370 (1983).

[5] S. S. McGaugh, F. Lelli, and J. M. Schombert, "Radial Acceleration Relation in Rotationally Supported Galaxies," *Physical Review Letters* **117**, 201101 (2016).

[6] P. Li, F. Lelli, S. McGaugh, J. Schombert, and M. S. Pawlowski, "A Comprehensive Catalog of Dark Matter Halo Models for SPARC Galaxies," *Astrophysical Journal Supplement Series* **247**, 31 (2020).